

Representation of complex processing logic

- Decision trees
- Decision tables

Decision Trees

- **Decision trees**
 - Edges of a decision tree represent conditions
 - Leaf nodes represent actions to be performed.
- **A decision tree gives a graphic view of,**
 - Logic involved in decision making
 - Corresponding actions taken.

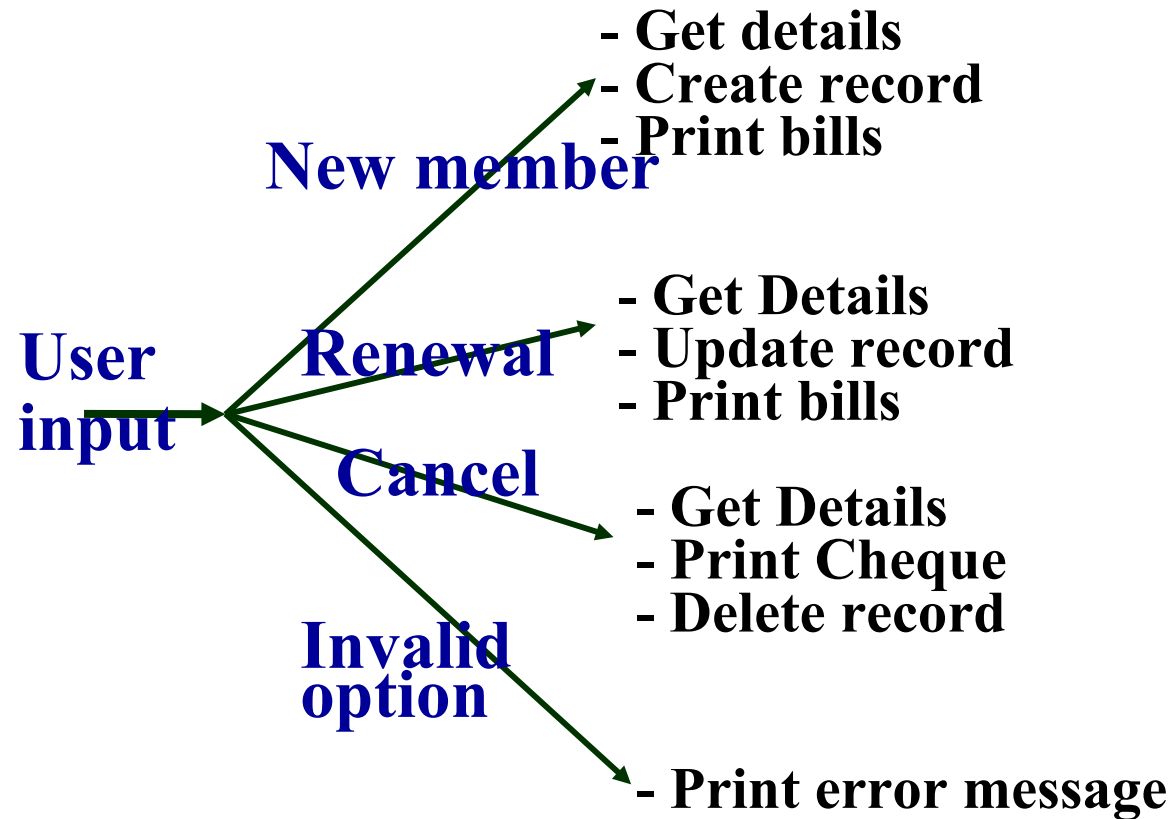
Example: Library Membership automation Software (LMS)

- A LMS should support following three options:
 - New member, Renewal, Cancel membership.
- When the new member option is selected,
 - The software asks details about the member name, address, phone number, etc.
- If proper information is entered,
 - A membership record for the member is created
 - A bill is printed for the annual membership charge plus the security deposit payable.

LMS Example (cont.)

- If the renewal option is chosen,
 - LMS asks the member's name and his membership number
 - checks whether he is a valid member.
 - If the name represents a valid member,
 - the membership expiry date is updated and the annual membership bill is printed,
 - Otherwise, an error message is displayed.
- If the cancel membership option is selected and the name of a valid member is entered,
 - The membership is cancelled,
 - A cheque for the balance amount due to the member is printed,
 - The membership record is deleted.

Decision Tree



Decision Table

- Decision tables specify:
 - Which variables are to be tested,
 - What actions are to be taken if the conditions are true,
 - The order in which decision making is performed.
- A decision table shows in a tabular form:
 - Processing logic and corresponding actions
- Upper rows of the table specify:
 - The variables or conditions to be evaluated
- Lower rows of the table specify:
 - The actions to be taken when the corresponding conditions are satisfied.

Decision Table

- In technical terminology,
 - a column of the table is called a rule
 - A rule implies:
 - if a condition is true, then execute the corresponding action.

LMS Example - Decision Table

- **Conditions**

Valid selection	NO	YES	YES	YES
New member	--	YES	NO	NO
Renewal	--	NO	YES	NO
Cancellation	--	NO	NO	YES

- **Actions**

Display error message
Ask member's name etc.
Build customer record
Generate bill
Ask membership details
Update expiry date
Print cheque
Delete record

Compare : Decision Tree vs. Decision Table

- Both decision tables and decision trees
 - Can represent complex program logic.
- **Decision trees** are easier to read and understand
 - When the **number of conditions are small**.
- **Decision tables** help to **look at every possible combination of conditions**.